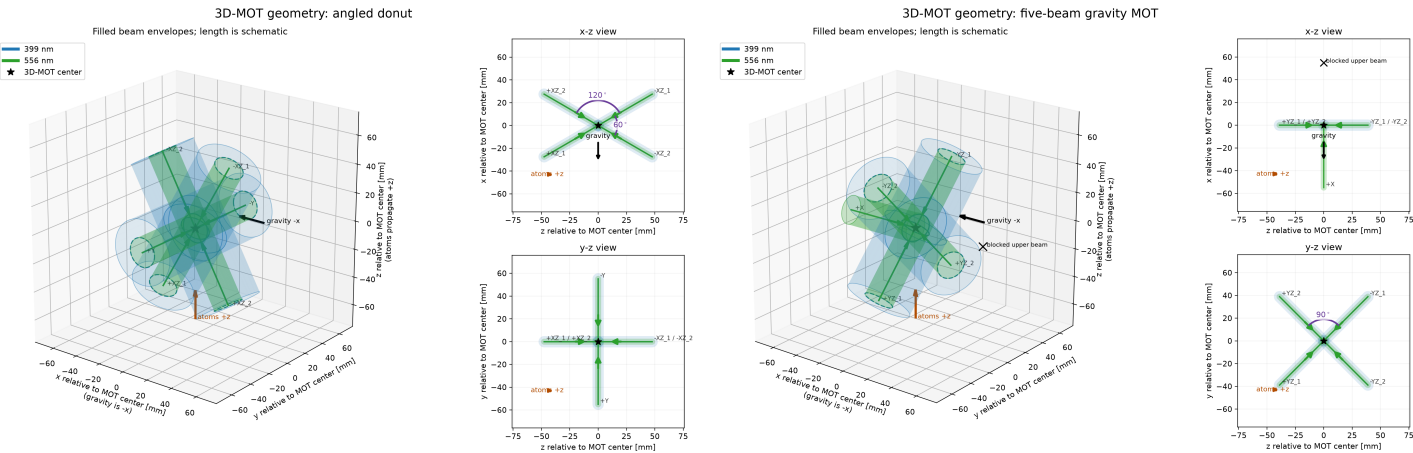


Selecting the two 3D-MOT candidates

Simulation evidence before the full optimization campaign

Decision: fully optimize the angled donut as the performance reference and the gravity-assisted five-beam MOT as the experimentally simpler alternative.

1. The candidates



Angled donut: six green core beams plus six coaxial, center-blocked blue beams. Highest capture, but more optics and alignment.

Five-beam MOT: one green counterbeam is replaced by gravity. Simpler to build, but presently lower capture and more asymmetric.

2. Why reduced-blue six-beam variants were not selected

Scope. Motivated by Plotkin-Swing et al. (2020), we screened two- and four-blue alternatives. These are not equal-budget global optimizations; screening asks which families show enough performance and experimental value to justify the expensive production search.

Configuration	Tested capture	Green	Blue	Main advantage	Main risk / cost	Decision
Full angled donut	12,992/15,840 (82.02%)	6	6	Highest demonstrated capture	Most optics and alignment	Optimize
Gravity-assisted five-beam	4,706/15,840 (29.71%)	5	4	Fewer beams; simpler implementation	Asymmetric balance; lower capture	Optimize
Four-blue entrance + backstop	4,629/15,840 (29.22%)	6	4	Avoids full blue shell	Six green beams; gate placement	Screen only

Five-beam versus four-blue: the observed difference is 77 atoms = 0.49 percentage points. A valid paired confidence interval cannot be reconstructed from the archived aggregate summary, so this difference is not claimed to be statistically meaningful. Five-beam is retained for comparable tested-point performance with fewer beams, not proven capture superiority.

What the full optimization will deliver

High-level plan, reporting metric, and uncertainty statement

Primary metric at 100 ms

An atom is usable when its distance from the MOT center is ≤ 5 mm and its total speed is ≤ 1 m/s. Earlier excursions do not permanently disqualify an atom.

Final result for each geometry

Recommended settings; mean conditional efficiency X with a 95% confidence interval; expected 3D-MOT atomic flux; and an empirically supported 7D or 10D operating box.

Data separation and statistical claim

Prevent selection bias

Discovery: a balanced 600-particle batch, 50 particles from each of 12 ensembles. Early checking and refinement use separate existing ensembles. The existing 15,840 survivors support candidate selection, not the unbiased final claim. Newly generated survivors stay sealed until settings and box construction are locked.

Uncertainty and flux

A crossed hierarchical bootstrap independently resamples 2D-MOT ensembles and recoil-seed realizations while preserving particle grouping within each ensemble. Report a 95% CI for mean X and new-ensemble variation separately. With X as a fraction: $\Phi_{3D} = X \Phi_{2D}$. Expected $\Phi_{2D} = 6.78e9$ atoms/s.

Bounded campaign budget and stopping rules

Stage	Max evaluations	Particles x seeds	Node-h	Elapsed / 3 nodes
Implementation	24 diagnostics	2-600 x 1	6	≤ 2 h
Discovery	350 donut + 600 five-beam	600 x 1	216	≤ 72 h
Refinement + finalists	16 + 3 / family	3,000 x 3; 15,840 x 3	504	≤ 168 h
Robust box	69 + 8 / family	1,800 x 1; 3,000 x 3	360	≤ 120 h
Checks + final report	12 / family + 2 nominal	600 x 3; ~16,000 x 3	216	≤ 72 h

Planning ceiling: 1,302 node-hours. One optional extension of at most 100 additional donut trials or 150 additional five-beam trials is permitted only after a stop-or-approve checkpoint. The 72-node-hour cap applies only if Stage-0 timing confirms the chosen block fits; otherwise reduce the block.

Approval requested

1. Approve the angled donut and five-beam MOT as the two production-search families.
2. Confirm or revise the provisional parameter bounds and laboratory constraints.
3. Approve the computational budget, bounded extension, and stopping rules.
4. Confirm whether a loss tolerance of $\delta = 1.0$ percentage point is experimentally negligible.
5. Confirm whether the unpaired lower green beam has independently controlled power, detuning, and waist.

Appendix A | Why the full donut is retained

Detailed evidence behind candidate selection

Common comparison: the same 600 validated 2D-MOT survivors, initial states, numerical settings, and recoil streams. Primary endpoint: usable at 100 ms (radius ≤ 5 mm and total speed ≤ 1 m/s). Earlier excursions do not disqualify atoms that return.

What the reduced-blue tests established

Full six-blue donut

491/600 (81.8%)

Removing the +z-component pair reduced capture to 65/600. Removing the transverse-y pair reduced it to 333/600.

Both the opposite longitudinal sense and transverse blue cooling therefore contribute substantially.

Two-blue single pass

Only one blue pair performs the main entrance slowing.

Initial: 5/600.
s0 + detuning tuned: 52/600.
s0 + detuning + position tuned: 115/600 (19.2%).

Optimization helps, but one encounter still accepts a narrow phase-space subset.

Four-blue finite gates

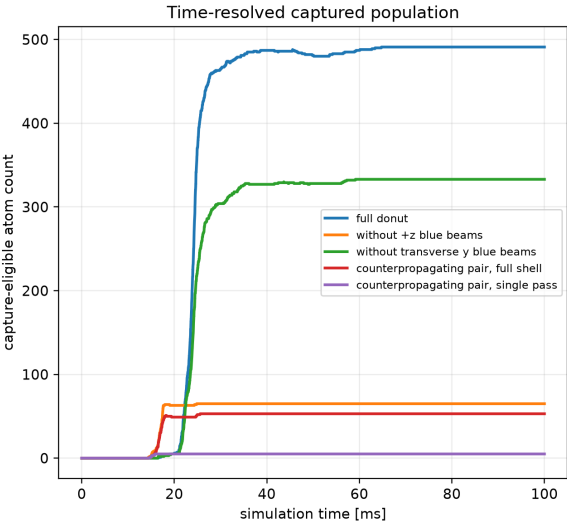
A second blue pair is placed downstream with the opposite role: it returns atoms that reverse or overshoot after the entrance slowing stage.

Best tested: 188/600 (31.3%).

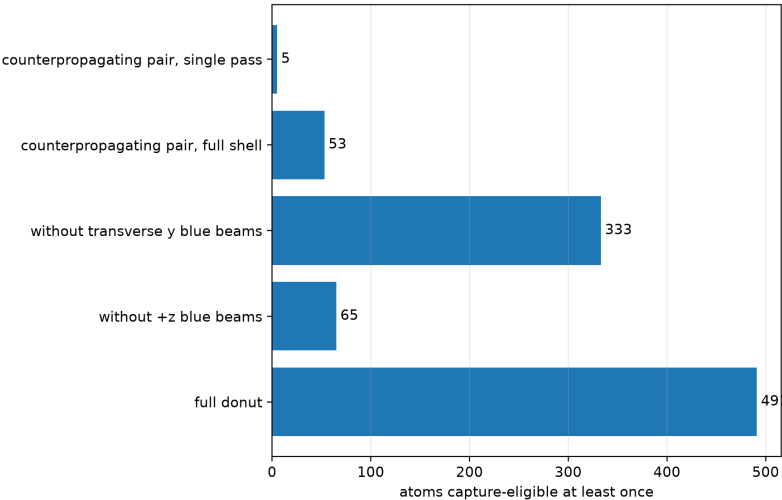
This confirms the value of the return pair, but it remains 303 atoms below the full donut.

Paired 600-particle ablation evidence

Which 399-nm beam functions make the donut effective?



Paired ablation outcome



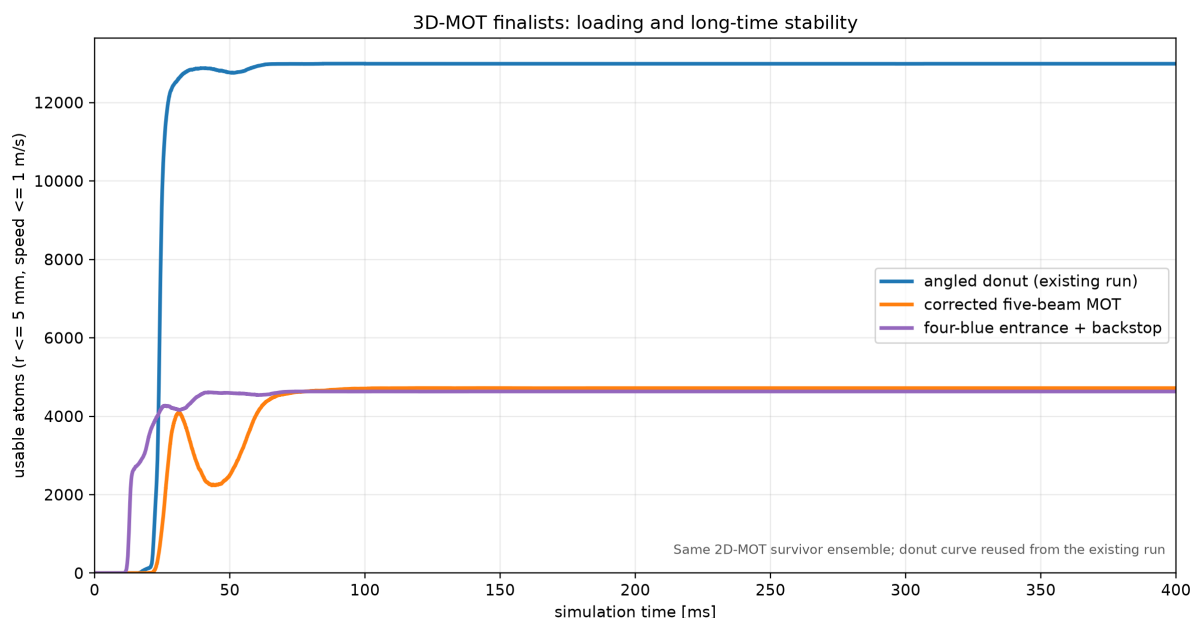
Reading: each curve uses the same initial particles and matched recoil seeds (the same seed index is applied to every configuration, reducing noise in configuration-to-configuration differences). Initial ablations and limited scans are screening evidence, not equal-budget global optimizations.

Five-beam tradeoff

Appendix B | Corrected geometry, pre-optimization evidence, and practical tradeoff

Architecture: one green counterbeam on the gravity axis is removed and gravity assists the remaining unpaired green beam. The other four green beams form two counterpropagating pairs with blue shells. Force-sign tests include gravity and verify restoring behavior on both sides of x , y , and z .

Direct full-ensemble loading and stability through 400 ms



Candidate-selection run on 15,840 shared survivors: donut 12,992 at 100 ms and 12,989 at 400 ms; five-beam 4,706 and 4,718; four-blue entrance + backstop 4,629 and 4,631. All are flat after loading. The five-beam/four-blue difference is 77 atoms (0.49 points); the archived aggregate summary cannot support a valid paired CI, so superiority is not claimed.

Decision tradeoff

Full angled donut

Advantages

- Highest capture by a large margin.
- Broad entrance-phase-space acceptance.
- Repeated cooling after reversals.
- Only 10 losses among 12,989 survivors from 100 to 400 ms in the tested hold.

Costs

- Six green and six blue beams.
- More optics, alignment, and access constraints.

Gravity-assisted five-beam MOT

Advantages

- One fewer green MOT beam.
- Simpler laboratory construction and alignment.
- Repeatable nonzero capture in the corrected geometry.
- No measurable population decay through 400 ms at the tested point.

Risks

- Present capture is much lower.
- Gravity is fixed, not a velocity-selective optical counterbeam.
- Greater sensitivity to force balance, gradient, and the unpaired beam is plausible.

Conclusion: pursue the donut as the performance reference and the five-beam as the simpler alternative, whose tested-point performance is comparable to the four-blue configuration while using fewer beams. These are screening decisions, not equal-budget rankings.

Appendix C | What is optimized

Variables, provisional domains, and laboratory constraints

Seven-dimensional angled-donut search

Six green core beams

1. s0
2. detuning [Gamma]
3. waist [m]

Six blue shell beams

4. s0
5. detuning [Gamma]
6. waist [m]

Magnetic field

7. gradient [G/cm]

Ten-dimensional five-beam search

Unpaired lower green: 1-3. s0, detuning, waist

Four paired green: 4-6. s0, detuning, waist

Four paired blue: 7-9. s0, detuning, waist

Magnetic field: 10. gradient [G/cm]

Production code must support independent lower-green detuning and waist before this search starts.

Planning domains - blue s0 cap confirmed; other bounds provisional

Laser groups

Green s0: 0.5-40
Green detuning: -35 to -5 Gamma
Green waist: 5-20 mm
Blue s0: 0.2-1.5 (hard upper limit)
Blue detuning: -6 to -0.5 Gamma
Blue waist: 10-25 mm

Gradients

Five-beam: 0.75-3 G/cm
Donut: 0.5-6 G/cm

Fixed geometry choices

Beam angles, polarizations, strong magnetic axis, cutoff geometry, MOT center, and gravity direction are not re-optimized.

Why domains remain provisional

The best tested five-beam point reached the low-gradient boundary, so the final range must extend below 1.4 G/cm. Earlier donut settings also came from boundary studies; neither search should be centered too narrowly.

Laboratory inputs required before production

1. Feasible minimum, maximum, adjustment resolution, calibration uncertainty, and drift for every s0, detuning, waist, and gradient.
2. Available optical power for each beam group. Since Gaussian-beam power scales approximately as $s0 \times \text{waist}^2$, s0 and waist cannot be optimized as an unconstrained rectangle.
3. Whether the unpaired green beam has independently controlled detuning and waist.
4. Whether beams within each group share common power, frequency, and waist controls.
5. The largest conditional-capture loss considered experimentally negligible for the robust parameter box.

Implementation gate: add per-group controls; verify force signs and exactly zero blue intensity in protected cores; reject infeasible power combinations; run 2-atom and 12-atom smoke tests; verify atomic checkpointing, restart, and live progress before allocating 24-hour jobs.

Appendix D | Search execution

Multi-fidelity Optuna, independent validation, and Zeus execution

Independent particle pools prevent optimistic reporting

Existing sample: 20 independent 2D-MOT ensembles, 15,840 survivors (756-840 each). Discovery uses 12 ensembles and a balanced 600-particle batch (50 per ensemble). Four separate ensembles support refinement; four support an early independent check. The 15,840 survivors are used for final candidate selection, not the unbiased performance claim. Newly generated final-validation ensembles remain sealed until all settings and the box procedure are locked.

Search stages

1. Broad discovery

Common balanced 600-particle set and one matched recoil seed. Hard caps: 350 valid donut trials and 600 valid five-beam trials. One optional extension: at most 100 additional donut or 150 additional five-beam trials, then stop.

2. Early generalization

At most 12 leading regions per family on an independent balanced 600-particle batch with 3 matched seeds. A region must pass before refinement.

3. Refinement

At most 16 points per family: 3,000 particles x 3 matched seeds. Rank by lower confidence bound; statistically tied points are ordered by robustness, interior margin, power, then sensitivity.

4. Finalists

At most 3 points per family on all 15,840 existing survivors x 3 seeds. Add seeds only for unresolved paired comparisons. Final newly generated ensembles remain untouched.

Zeus execution and restart safety

Three long-lived workers: 3 nodes x 200 cores, 64 GB per node, 24-hour walltime. Planning ceiling: 1,302 node-hours before extension, including 6 node-hours for implementation/smoke tests. The optional extension is capped at 72 node-hours and proceeds only if measured Stage-0 throughput shows the chosen block fits; otherwise reduce the block. Save every trial atomically. Constraint violations are PRUNED_INFEASIBLE; numerical/time/output failures are FAIL.

Final claims

Appendix E | Robust parameter box, uncertainty, flux propagation, and deliverables

Near-optimal 7D or 10D hyperrectangle

Construction: fit a local surrogate to a space-filling design around a robust nominal point; snap bounds to laboratory resolution; shrink the box until its risk-directed validation points pass $\delta = 1.0$ percentage point, subject to approval.

Finite design: center and all face midpoints (14 or 20); at most 16 fractional/risk-directed corners; at most 24 interior/boundary points; at most 8 surrogate-adversarial points; high-statistics reevaluation of at most 8 worst points. Use one-sided max-T bootstrap simultaneous upper bounds across that finite high-statistics set.

Claim: the box is an empirically supported operating region. Behavior between tested points is assessed by the surrogate and adversarial sampling, not guaranteed mathematically.

Final prediction and uncertainty

Conditional efficiency

X = mean usable 3D-MOT fraction evaluated only on newly generated sealed 2D-MOT survivor ensembles. The existing 15,840 survivors select candidates; they do not support the unbiased final claim.

Crossed hierarchical bootstrap: independently resample ensemble clusters and recoil-seed realizations, preserving particle grouping within each ensemble. Report the 95% CI for mean X; report new-ensemble variation separately.

Mean-CI half-width target: ≤ 0.75 percentage points.

End-to-end atomic flux

Modeled Yb-171 oven flux: 7.39×10^{13} atoms/s.

Expected 2D-MOT flux: 6.78×10^9 atoms/s.

95% interval: $(6.64-6.93) \times 10^9$ atoms/s.

With X defined as a fraction: $\Phi_{3D} = X \Phi_{2D}$. Propagate uncertainty from both stages. Conditional 3D efficiency and oven-to-3D efficiency remain separate because their denominators differ.

Final package delivered for each geometry

Operating recommendation

- Nominal parameter values and required optical power.
- Mean conditional capture X with its 95% confidence interval and predicted usable atomic flux.
- Empirically supported near-optimal hyperrectangle and declared δ .

Physics diagnostics

- Population versus time through 100 ms.
- Longer retention and τ only when a genuine exponential tail passes loss and fit-quality checks.
- Final survivor checkpoints, common-scale phase-space acceptance, and representative captured and failed trajectories.

Decision support

- Capture, robustness, power, beam count, alignment complexity, optical access, and implementation risk shown side by side. The simulation quantifies the tradeoff; the laboratory team decides how much atomic flux is worth the simpler geometry.